

Direct Numerical Simulations of isotropic turbulence interacting with a shock wave

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1 Introduction

The interaction of shock waves with turbulence occurs in many areas, including astrophysics, inertial confinement fusion, and internal and external flows in hypersonic flight and propulsion. In such flows, the coexistence of shock waves with some background turbulence is an almost unavoidable feature that generates complicating effects even when the shocks are relatively weak and a detailed understanding remains far from reach. For this reason, we focus on the interaction between isotropic turbulence and a normal shock wave, as one of the basic unit problems to study these phenomena.

In many practical applications, the shock width is much thinner than the turbulence scales, while the turbulent Mach number, M_t , is relatively modest. The limiting case of small M_t can be addressed analytically, if one further assumes negligible viscous effects. This implies large Reynolds numbers, Re , which is usually the case in practical flows. This approach is known as the Linear Interaction Analysis (LIA, [1, 2]). In this limit, the shock is assumed to be a perturbed discontinuity and the linearized Euler equations and Rankine-Hugoniot jump conditions are used to derive analytical solutions.

Since the LIA was introduced in 1950s, a number of studies have presented comparisons between the LIA and numerical simulations. Due to the high cost necessary to resolve all the turbulence scales as well as the shock structure, previous studies using Direct Numerical Simulations (DNS) [3, 4] could consider weak shocks only ($M_s \leq 1.5$, where M_s is the shock Mach number) at parameter ranges where the interaction was dominated by viscous and/or nonlinear effects and, consequently, showed limited agreement with the LIA solutions. More recently, using shock-capturing schemes, the range of M_s values was extended considerably [5, 6]. The comparison with the LIA was made after the viscous effects behind the shock were compensated. A good agreement was found for the turbulent kinetic energy, but in-

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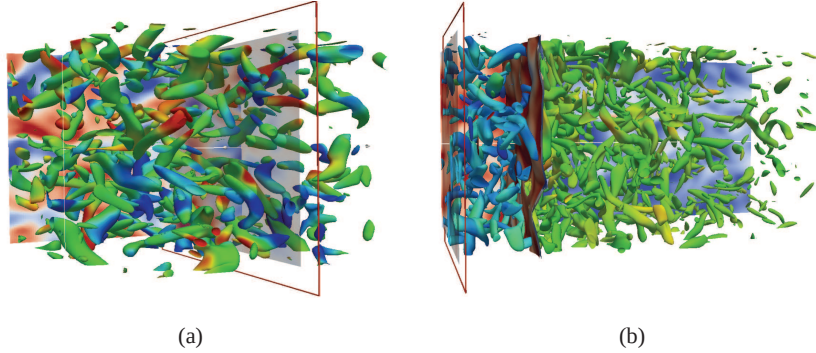


Fig. 1 Numerical set-up: data recorded from forced isotropic turbulence simulations (a) are fed through the inlet of a shock tube (b). The red rectangle is the plane where the flow data are recorded (a) and the location of inlet feeding (b). Eddy structures are visualized by the Q-criterion ($Q = 0.0125c_1^2k_0^2$). The shock location is identified through the isosurface with dilatation $= -0.5 c_1 k_0$.

dividual Reynolds stresses did not match the LIA prediction. Thus, as a result of the limited agreement presented in previous studies, the significance of the LIA has not yet been fully appreciated.

This study aims to investigate and clarify the role of the LIA as a limiting case and emphasize the importance in many practical applications when M_t is small even if the Reynolds number is not very large. We have conducted fully resolved simulations covering the parameter space from linear inviscid, close to the LIA limit, to regimes dominated by nonlinear and/or viscous effects. The M_s values considered are 1.1, 1.2, 1.4, 1.8, and 2.2. The simulations, described below, show that the DNS results do converge to the LIA solution as nonlinear effects become negligible. In addition, the consistent coverage of the parameter space reveals that the streamwise velocity amplification does not collapse onto a single parameter, as proposed in Ref. [7].

2 Numerical Method

The compressible Navier-Stokes equations with the perfect gas assumption are solved using the CFDNS code [8, 9]. The ratio of specific heats is $\gamma = 1.4$, the viscosity varies with the temperature as $\mu = \mu_0(T/T_0)^{0.75}$, and the Prandtl number is $Pr = 0.7$. The flow variables are nondimensionalized by the upstream mean density, temperature, and speed of sound, c_1 . The spatial discretization is performed using sixth-order compact finite differences and the variable time step Runge-Kutta-Fehlberg (RK45) method is used for time advancement. The mesh sizes are large enough such that all flow scales (including the shock width) are accurately resolved without applying any turbulence model, shock-capturing, or filtering methods and

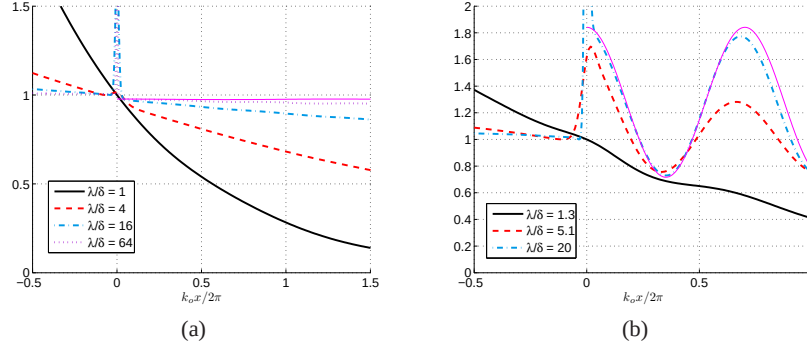


Fig. 2 The streamwise Reynolds stress, $R_{11} = \langle u'u' \rangle$, variation through the shock for a single vortical plane-wave with various wavelengths, λ , from DNS and LIA, at $M_s = 1.2$. The angle between the wave vector and the streamwise direction is $\psi = 45^\circ$ ((a)) and 75° ((b)). The latter is close to the critical angle ($\psi \simeq 70^\circ$). The top magenta solid line represents the LIA solution.

the results of the simulations are converged under grid refinement. Depending on the target flow state, 64 to 256 grid points are used in the transverse directions and 512 to 4096 grid points, together with a non-uniform mesh which is finest around the shock, are used in the streamwise direction. An accelerating layer $\sim 10\%$ of the streamwise domain length is used at the outflow boundary [10] to ensure non-reflecting boundary conditions.

In order to provide realistic turbulence upstream of the shock wave, auxiliary forced isotropic turbulence simulations are performed with prescribed background velocity matching the shock speed (Fig. 1 (a)). The statistically stationary state is reached using the linear forcing method [9], which has the advantage of specifying the Kolmogorov length scale, η , the ratio of dilatational to solenoidal kinetic energies, χ , and most energetic wavenumber, k_0 , at the outset. Here, these values are $\eta/dy = 1.1 \sim 2.0$, $\chi = 0.0005 \sim 0.12$, and $k_0 = 4$. Plane data are recorded at a fixed plane perpendicular to the streamwise direction and the data are fed through the inlet of a shock-tube with a stationary shock (Fig. 1 (b)). This method avoids limitations of using the Taylor hypothesis which is required to transform temporally decaying isotropic turbulence data into spatially decaying turbulence.

3 Results

For all simulations, the turbulence statistics are collected after one flow-through time to remove the initial transients and the averages are taken over the transverse directions and time. More than three flow-through times are used to collect instantaneous data and the results are converged. In all figures, the mean location of the shock is at streamwise position $x = 0$ and the turbulence quantities are non-dimensionalized by their values immediately upstream of the shock.

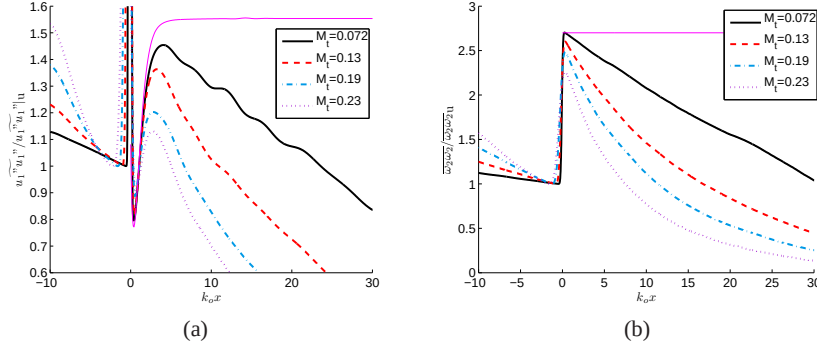


Fig. 3 Convergence of R_{11} ((a)) and transverse vorticity variance, Ω_{tr} ((b)) variation through a $M_s = 1.4$ shock to the LIA solution as the nonlinear and viscous effects become small. All the DNS runs have $Re_\lambda \simeq 10$; however, δ/η decreases with M_t . The correspondent values of this ratio are $\delta/\eta = 1.5, 1.2, 0.81$, and 0.42 , as M_t decreases from 0.23 to 0.072 . LIA predictions are shown with magenta lines.

When there is a large separation in scale between the shock width and the incoming small amplitude disturbances, viscous and nonlinear effects become negligible during the interaction process. In this case, the DNS results should be close to the LIA prediction. Fig. 2 compares the variation of R_{11} through the shock, for a single vortical plane-wave with small amplitude between DNS and LIA. Note that the LIA solution is independent on the wavelength λ and the amplitude of the perturbation and only depends on ψ and M_s . When the ratio between λ and the laminar shock thickness δ is small, the DNS results are very different than the LIA solution, with the limiting case, $\lambda/\delta = 1$, showing no amplification at all. However, as λ/δ increases, the DNS results converge to the LIA prediction, even close to the critical angle (here, the LIA amplification is large, requiring stronger constraints to reduce the nonlinear effects in DNS). Thus, scale separation can be a criterion for controlling the viscous effects on the interaction.

The convergence to the LIA prediction is shown for full turbulent fields in Fig. 3. Here, the scale separation can be controlled by the ratio δ/η . This ratio was also presented as a scale criterion for using shock capturing schemes [11]. The DNS amplifications converge to the LIA solutions when δ/η becomes small. Note that full convergence for R_{11} is not obtained at the smallest $\delta/\eta = 0.42$, unlike Ω_{tr} . The peak location of Ω_{tr} is immediately behind the shock wave. However, the peak of R_{11} is located at around one most energetic wavelength behind the shock and is affected by viscous effects after the shock interaction [5] when Re_λ is small. These effects are also minimized as δ/η and M_t decrease, since the eddy turnover time, and consequently, the decay distance increase. Nevertheless, the viscous effects behind the shock lead to slower R_{11} convergence to the LIA solution compared to Ω_{tr} .

Fig. 4 (a) shows the convergence of the DNS R_{11} amplifications to the LIA solutions, as M_t decreases, for all M_s values considered. The amplification is computed as the ratio of the peak value behind the shock to the value immediately upstream

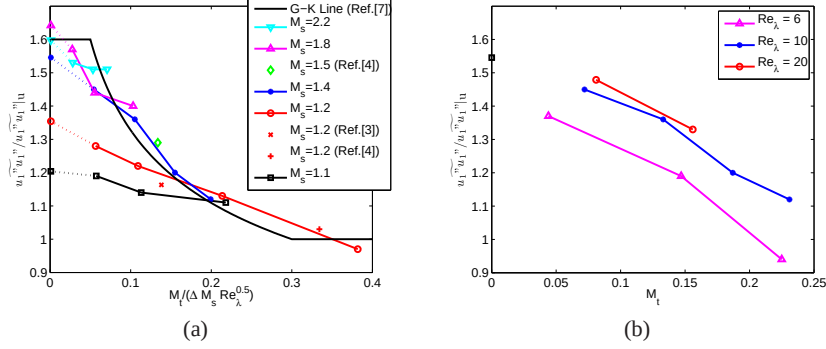


Fig. 4 (a) R_{11} as a function of $M_t / ((M_s - 1)\sqrt{Re_\lambda})$ which is $\sim \delta/\eta$ for different M_s and $Re_\lambda \simeq 10$. Symbols along the vertical axis represent LIA solution with the shape and color matched for the symbol-lines of corresponding M_s . LIA solution and corresponding DNS results are connected with dotted lines. For comparison, previous DNS results are also shown in from Ref. [3] and [4]. The black curve represents the amplification model from Ref. [7]. (b) R_{11} as a function of M_t for different Re_λ and $M_s = 1.4$. A symbol along the vertical axis represents LIA solution for $M_s = 1.4$.

of the shock, consistent to the LIA procedure. Though not presented here, the amplification of R_{22} measured at the same locations also shows similar convergence. Note that the results correspond to relatively low Re_λ . As most of the turbulence scales are much larger than δ , the viscous effects through the shock easily become negligible, and the results are not far from the LIA solution. Thus, it is stressed that even for quite viscous flows, the amplification can be predicted by the LIA provided that M_t is small enough to satisfy the linear approximation. As expected, the convergence rate increases with Re_λ (fig. 4 b). This aspect is being further examined with ongoing higher resolution simulations.

The ratio δ/η can be written as $\delta/\eta \simeq 7.69 M_t / (Re_\lambda^{0.5} (M_s - 1))$, which combines the effects of M_t , M_s , and Re_λ . This ratio was proposed in Ref. [7] as a universal parameter which characterizes the turbulence amplification. The proposal was based on the apparent collapse of previous data for R_{11} amplification onto a single curve as a function of δ/η (black line in Fig. 4 a). Thus, due to 1) the relative limited coverage of M_t values for a given M_s and 2) overlooking the acoustic modes in the incoming turbulence in the previous studies, it may lead to the impression that such a parameter can be derived. Although the inlet turbulence is usually generated with vortical modes only, if M_t is not very small, non-negligible acoustic modes can develop ahead of the shock. For high M_s values ($M_s > 2.5$), significantly larger amplification of the acoustic modes is predicted by the LIA. For example, this is 10 times larger than that of vortical modes for $M_s = 5$. Even for $M_t = 0.2$, estimating $\chi \sim M_t^2$, this leads to a non-negligible correction required to the LIA solution. Thus, one needs to include the correct amount of dilatational energy in the LIA solution for appropriate comparisons. The present simulations converge to the LIA vortical solution as M_t becomes small and exhibit the associated M_s dependence (Fig. 4 a).

4 Summary and Conclusions

The interaction between isotropic turbulence and a normal shock wave is studied using Direct Numerical Simulations, with all flow scales (including the shock width) accurately solved. Realistic turbulence is generated in separate stationary isotropic simulations, with background velocity matching the shock speed, to avoid the use of the Taylor hypothesis. M_s ranges from 1.1 to 2.2. In this study, the vortical mode dominates upstream of the shock and the simulations cover the parameter space from linear inviscid, close to the LIA limit, to regimes dominated by nonlinear and/or viscous effects. It is shown, for the first time, that the DNS results do converge to the LIA solution as viscous and nonlinear effects become negligible. Since in many practical applications M_t is small and the Reynolds number is large, this emphasizes the importance of the LIA beyond that of a limiting case. In addition, the results reveal that the streamwise velocity amplification does not collapse onto a single parameter (i.e. δ/η), as proposed in some previous studies. Higher resolution simulations are underway and will be discussed elsewhere.

Los Alamos National Laboratory is operated by Los Alamos National Security, LLC for the US Department of Energy NNSA under contract no. DE-AC52-06NA25396. Computational resources were provided by LANL Institutional Computing (IC) Program and Sequoia Open Science Program at Lawrence Livermore National Laboratory.

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